

ITMO-453 Threat Model

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Executive Summary

High level system description

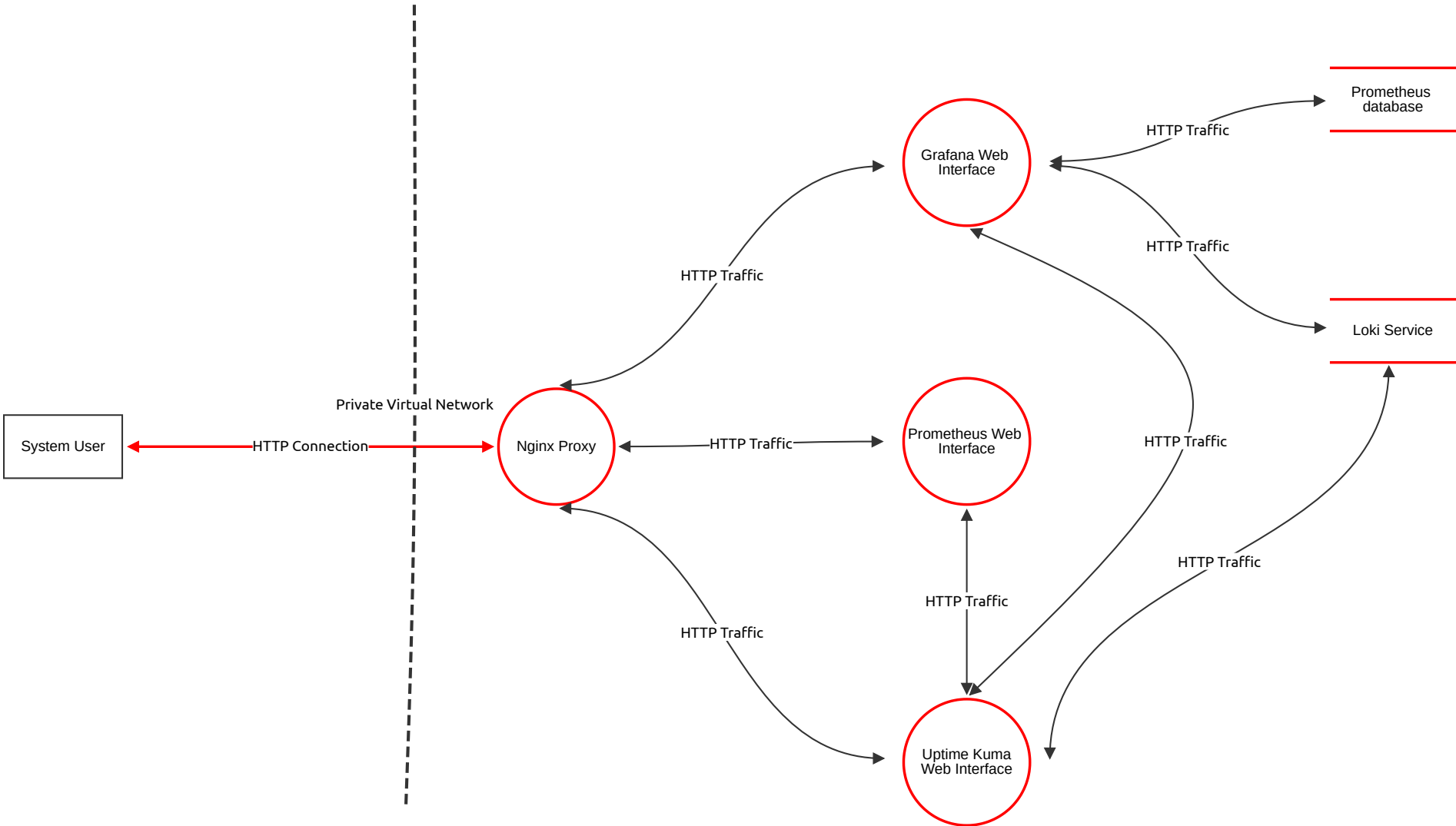
An ubuntu 24.04 server running in virtualbox with several web services installed.

Summary

Total Threats	8
Total Mitigated	0
Total Open	8
Open / Critical Severity	0
Open / High Severity	3
Open / Medium Severity	3
Open / Low Severity	2

Ubuntu Server STRIDE Diagram

A STRIDE diagram for an ubuntu server outlining general system architecture and potential threats.



Ubuntu Server STRIDE Diagram

Nginx Proxy (Process)

Description: An instance of nginx acting as a simple server and proxy

Number	Title	Type	Severity	Status	Score	Description	Mitigations
2	OS and Service Fingerprinting	Information disclosure	High	Open	5	An attacker could scan the system using tools such as nmap and discover information about running services and OS info.	Host services through a proxy and configure the system to minimize the amount of information that is revealed. Only open firewall ports that are absolutely necessary.
4	DDOS Attack	Denial of service	High	Open	5	An attacker could execute a denial of service attack and bring down the system.	Implement protective measures such as fail2ban or cloudflare.

Grafana Web Interface (Process)

Description: Web interface for Grafana.

Number	Title	Type	Severity	Status	Score	Description	Mitigations
1	Unauthorized Login	Elevation of privilege	Medium	Open	3	An attacker could take advantage of a weak password and gain access to an administrators dashboard.	Implement a strong organisational password policy.

Prometheus Web Interface (Process)

Description: Web interface for prometheus.

Number	Title	Type	Severity	Status	Score	Description	Mitigations
1	Unauthorized Login	Elevation of privilege	Medium	Open	3	An attacker could take advantage of a weak password and gain access to an administrators dashboard.	Implement a strong organisational password policy.

Uptime Kuma Web Interface (Process)

Description: Web interface for uptime-kuma. Installed inside of a docker container.

Number	Title	Type	Severity	Status	Score	Description	Mitigations
1	Unauthorized Login	Elevation of privilege	Medium	Open	3	An attacker could take advantage of a weak password and gain access to an administrators dashboard.	Implement a strong organisational password policy.

Prometheus database (Store)

Description: Time series database for prometheus.

Number	Title	Type	Severity	Status	Score	Description	Mitigations
6	Log contains sensitive data	Information disclosure	Low	Open	2	Logs are read by unauthorised users or made public, sensitive data is then disclosed	Minimise any sensitive data contained in logs, consider encryption techniques

Loki Service (Store)

Description: Loki service for collecting system logs

Number	Title	Type	Severity	Status	Score	Description	Mitigations
5	Log contains sensitive data	Information disclosure	Low	Open	2	Logs are read by unauthorised users or made public, sensitive data is then disclosed	Minimise any sensitive data contained in logs, consider encryption techniques

System User (Actor)

Description: A generic user for the system.

Number	Title	Type	Severity	Status	Score	Description	Mitigations
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HTTP Connection (Data Flow)

Description: HTTP Communication between the user and the nginx proxy.

Number	Title	Type	Severity	Status	Score	Description	Mitigations
3	Eavesdropping	Tampering	High	Open	5	An attacker could listen in on http traffic over a public connection	Implement HTTPS on the system.

HTTP Traffic (Data Flow)

Description: Internal HTTP traffic between services

Number	Title	Type	Severity	Status	Score	Description	Mitigations
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